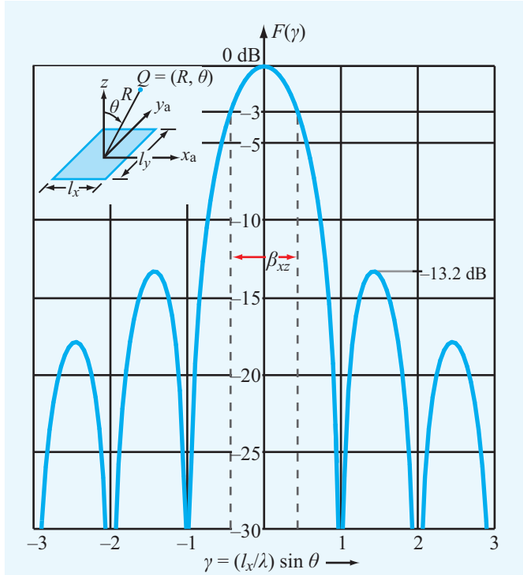


Figure 9-23 Normalized radiation pattern of a uniformly illuminated rectangular aperture in the x–z plane ($\phi = 0$).

► The sinc function is maximum when its argument is zero; $\text{sinc}(0) = 1$. ◄

This occurs when $\theta = 0$. Hence, at a fixed range R, $S_{\max} = S(\theta = 0) = S_0$. The normalized radiation intensity is then given by

$$F(\theta) = \frac{S(R,\theta)}{S_{\max}} = \text{sinc}^2(\pi l_x \sin\theta/\lambda) = \text{sinc}^2(\pi\gamma) \qquad (x\text{–}z \text{ plane}). \tag{9.84}$$

Figure 9-23 shows $F(\theta)$ plotted (on a decibel scale) as a function of the intermediate variable $\gamma = (l_x/\lambda)\sin\theta$. The pattern exhibits nulls at nonzero integer values of γ.

9-8.1 Beamwidth

The normalized radiation intensity $F(\theta)$ is symmetrical in the x–z plane, and its maximum is along the boresight direction ($\theta = 0$, in this case). Its half-power beamwidth $\beta_{xz} = \theta_2 - \theta_1$, where θ_1 and θ_2 are the values of θ at which $F(\theta, 0) = 0.5$ (or −3 dB on a decibel scale), as shown in **Fig. 9-23**. Since the pattern is symmetrical with respect to $\theta = 0$, $\theta_1 = -\theta_2$, and $\beta_{xz} = 2\theta_2$. The angle θ_2 can be obtained from a solution of

$$F(\theta_2) = \text{sinc}^2(\pi l_x \sin\theta/\lambda) = 0.5. \tag{9.85}$$

From tabulated values of the sinc function, it is found that Eq. (9.85) yields the result

$$\frac{\pi l_x}{\lambda}\sin\theta_2 = 1.39, \tag{9.86}$$

or

$$\sin\theta_2 = 0.44\frac{\lambda}{l_x}. \tag{9.87}$$

Because $\lambda/l_x \ll 1$ (a fundamental condition of scalar diffraction theory is that the aperture dimensions be much larger than the wavelength λ), θ_2 is a small angle, in which case we can use the approximation $\sin\theta_2 \approx \theta_2$. Hence,

$$\beta_{xz} = 2\theta_2 \approx 2\sin\theta_2 = 0.88\frac{\lambda}{l_x} \quad \text{(rad)}. \tag{9.88a}$$

A similar solution for the y–z plane ($\phi = \pi/2$) gives

$$\beta_{yz} = 0.88\frac{\lambda}{l_y} \quad \text{(rad)}. \tag{9.88b}$$

► The uniform aperture distribution ($\tilde{E}_a = E_0$ across the aperture) gives a far-field pattern with the narrowest possible beamwidth. ◄

The first sidelobe level is 13.2 dB below the peak value (see **Fig. 9-23**), which is equivalent to 4.8% of the peak value. If the intended application calls for a pattern with a lower sidelobe level (to avoid interference with signals from sources along directions outside the main beam of the antenna pattern), this can be accomplished by using a ***tapered aperture distribution***—one that is a maximum at the center of the aperture and decreases toward the edges.

► A tapered distribution provides a pattern with lower side lobes, but the main lobe becomes wider. ◄

The steeper the taper, the lower are the side lobes and the wider is the main lobe. In general, the beamwidth in a given plane,

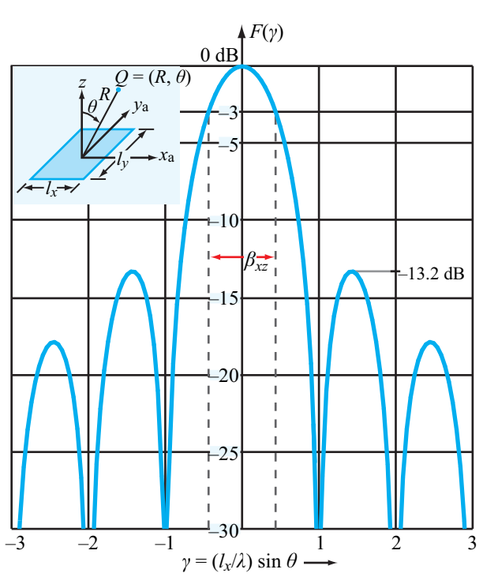


图 9-23 x-z 平面 ($\phi = 0$) 中均匀照明的矩形孔径的归一化辐射方向图。

► 当参数为零时，sinc 函数最大；正弦(0) = 1. ◄

这发生在 $\theta = 0$ 时。因此，在固定范围 R，$S_{max} = S(\theta = 0) = S_0$ 处。归一化辐射强度由下式给出

$$F(\theta) = \frac{S(R,\theta)}{S_{\max}} = \text{sinc}^2(\pi l_x \sin\theta/\lambda) = \text{sinc}^2(\pi\gamma) \qquad (x\text{–}z \text{ plane}). \tag{9.84}$$

图 9-23 显示了 F(θ)（以分贝刻度）作为中间变量 $\gamma = (l_x/\lambda)\sin\theta$ 的函数。该模式在 γ 的非零整数值处呈现空值。

9-8.1 波束宽度

归一化辐射强度 F(θ) 在 x-z 平面上对称，其最大值沿视轴方向（在本例中为 θ = 0）。其半功率波束宽度 $\beta_{xz} = \theta_2 - \theta_1$，其中 θ_1 和 θ_2 是 θ 的值，其中 F(θ, 0) = 0.5（或

−3 dB（分贝级），如图 9-23 所示。由于该图案相对于 $\theta = 0$、$\theta_1 = -\theta_2$ 和 $\beta_{xz} = 2\theta_2$ 对称。角度 θ_2 可以从以下解中获得

$$F(\theta_2) = \text{sinc}^2(\pi l_x \sin\theta/\lambda) = 0.5. \tag{9.85}$$

从 sinc 函数的表格值可以发现，方程：(9.85) 得出结果

$$\frac{\pi l_x}{\lambda}\sin\theta_2 = 1.39, \tag{9.86}$$

或者

$$\sin\theta_2 = 0.44\frac{\lambda}{l_x}. \tag{9.87}$$

由于$\lambda/l_x \ll 1$（标量衍射理论的基本条件是孔径尺寸远大于波长λ），θ_2是一个小角度，在这种情况下我们可以使用近似$\sin\theta_2 \approx \theta_2$。因此，

$$\beta_{xz} = 2\theta_2 \approx 2\sin\theta_2 = 0.88\frac{\lambda}{l_x} \quad \text{(rad)}. \tag{9.88a}$$

y–z 平面 ($\phi = \pi/2$) 的类似解给出

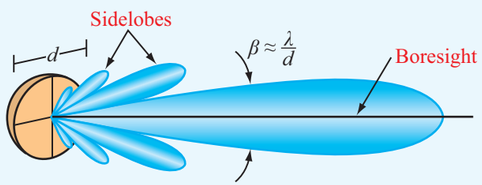
$$\beta_{yz} = 0.88\frac{\lambda}{l_y} \quad \text{(rad)}. \tag{9.88b}$$

► 均匀的孔径分布（孔径上的 $\tilde{E}_a = E_0$）提供了具有尽可能窄的波束宽度的远场方向图。 ◄

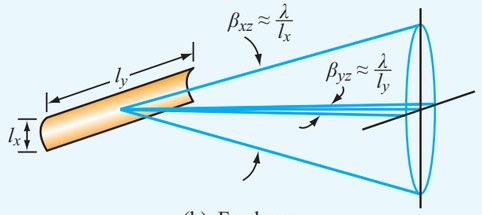
第一旁瓣电平比峰值低 13.2 dB（见图 9-23），相当于峰值的 4.8%。如果预期应用需要具有较低旁瓣电平的方向图（以避免沿天线方向图主波束之外的方向对来自源的信号产生干扰），则可以通过使用锥形孔径分布来实现，即孔径中心处最大值并向边缘减小。

► 锥形分布提供了具有较低旁瓣的模式，但主瓣变得更宽。 ◄

锥度越陡，旁瓣越低，主瓣越宽。一般来说，给定平面上的波束宽度，



(a) Pencil beam



(b) Fan beam

Figure 9-24 Radiation patterns of (a) a circular reflector and (b) a cylindrical reflector (side lobes not shown).

say the x–z plane, is given by

$$\beta_{xz} = k_x \frac{\lambda}{l_x}, \tag{9.89}$$

where k_x is a constant related to the steepness of the taper. For a uniform distribution with no taper, $k_x = 0.88$, and for a highly tapered distribution, $k_x \approx 2$. In the typical case, $k_x \approx 1$.

To illustrate the relationship between the antenna dimensions and the corresponding beam shape, we show in **Fig. 9-24** the radiation patterns of a circular reflector and a cylindrical reflector. The circular reflector has a circularly symmetric pattern, whereas the pattern of the cylindrical reflector has a narrow beam in the azimuth plane corresponding to its long dimension and a wide beam in the elevation plane corresponding to its narrow dimension. For a circularly symmetric antenna pattern, the beamwidth β is related to the diameter d by the approximate relation $\beta \approx \lambda/d$.

9-8.2 Directivity and Effective Area

In Section 9-2.3, we derived an approximate expression [Eq. (9.26)] for the antenna directivity D in terms of the half-power beamwidths β_{xz} and β_{yz} for antennas characterized by a single major lobe whose boresight is along the z direction:

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \tag{9.90}$$

If we use the approximate relations $\beta_{xz} \approx \lambda/l_x$ and $\beta_{yz} \approx \lambda/l_y$, we obtain

$$D \approx \frac{4\pi l_x l_y}{\lambda^2} = \frac{4\pi A_p}{\lambda^2}. \tag{9.91}$$

For any antenna, its directivity is related to its effective area A_e by Eq. (9.64):

$$D = \frac{4\pi A_e}{\lambda^2}. \tag{9.92}$$

► For aperture antennas, their effective apertures are approximately equal to their physical apertures; that is, $A_e \approx A_p$. ◄

Exercise 9-11: Verify that Eq. (9.86) is a solution of Eq. (9.85) by calculating $\text{sinc}^2 t$ for $t = 1.39$.

Exercise 9-12: With its boresight direction along z, a square aperture was observed to have half-power beamwidths of 3° in both the x–z and y–z planes. Determine its directivity in decibels.

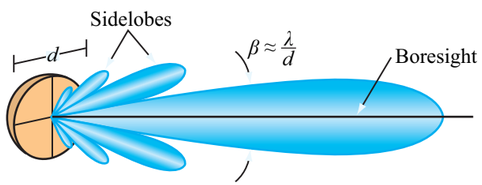
Answer: $D = 4{,}583.66 = 36.61$ dB. (See EM.)

Exercise 9-13: What condition must be satisfied in order to use scalar diffraction to compute the field radiated by an aperture antenna? Can we use it to compute the directional pattern of the eye's pupil ($d \approx 0.2$ cm) in the visible part of the spectrum ($\lambda = 0.35$ to $0.7\ \mu$m)? What would the beamwidth of the eye's directional pattern be at $\lambda = 0.5\ \mu$m?

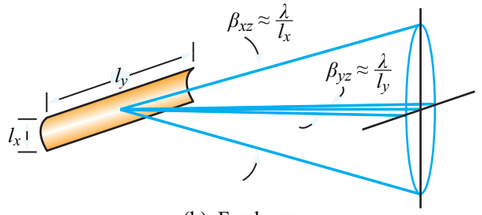
Answer: $\beta \approx \lambda/d = 2.5 \times 10^{-4}$ rad $= 0.86'$ (arc minute, with $60' = 1°$). (See EM.)

9-9 Antenna Arrays

AM broadcast services operate in the 535 to 1605 kHz band. The antennas they use are vertical dipoles mounted along tall towers. The antennas range in height from $\lambda/6$ to $5\lambda/8$, depending on the operating characteristics desired and other considerations. Their physical heights vary from 46 m (150 ft) to 274 m (900 ft); the wavelength at 1 MHz, approximately in the middle of the AM band, is 300 m. Because the field radiated by a single dipole is uniform in the horizontal plane



(a) Pencil beam



(b) Fan beam

图 9-24 (a) 圆形反射器和 (b) 圆柱形反射器的辐射方向图（未显示旁瓣）。

说 x–z 平面，由下式给出

$$\beta_{xz} = k_x \frac{\lambda}{l_x}, \tag{9.89}$$

其中 k_x 是与锥度陡度相关的常数。对于无锥度的均匀分布，$k_x = 0.88$；对于高锥度分布，$k_x \approx 2$。在典型情况下，$k_x \approx 1$。

为了说明天线尺寸和相应波束形状之间的关系，我们在图 9-24 中显示了圆形反射器和圆柱形反射器的辐射方向图。圆形反射器具有圆形对称图案，而圆柱形反射器的图案在与其长尺寸对应的方位平面中具有窄光束，在与其窄尺寸对应的仰角平面中具有宽光束。对于圆形对称天线方向图，波束宽度 β 与直径 d 通过近似关系 $\beta \approx \lambda/d$ 相关。

9-8.2 方向性和有效面积

在9-2.3节中，我们推导出了一个近似表达式[Eq. (9.26)]对于天线方向性 D，以半功率波束宽度 β_{xz} 和 β_{yz} 表示，天线特征为

视轴沿 z 方向的单主瓣：

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \tag{9.90}$$

如果我们使用近似关系 $\beta_{xz} \approx \lambda/l_x$ 和 $\beta_{yz} \approx \lambda/l_y$，我们得到

$$D \approx \frac{4\pi l_x l_y}{\lambda^2} = \frac{4\pi A_p}{\lambda^2}. \tag{9.91}$$

对于任何天线，其方向性与其有效面积A_e有关，如式(1)所示。 (9.64) ：

$$D = \frac{4\pi A_e}{\lambda^2}. \tag{9.92}$$

► 对于孔径天线，其有效孔径约等于其物理孔径；即 $A_e \approx A_p$。 ◄

练习 9-11：验证方程(9.86) 是方程 (9.86) 的解。 (9.85) 通过计算 $t = 1.39$ 的 $\text{sinc}^2 t$。

练习 9-12：当视轴方向沿 z 方向时，观察到方形孔径在 x-z 和 y-z 平面上的半功率波束宽度均为 3°。确定其方向性（以分贝为单位）。

Answer: $D = 4{,}583.66 = 36.61$ dB. (See EM.)

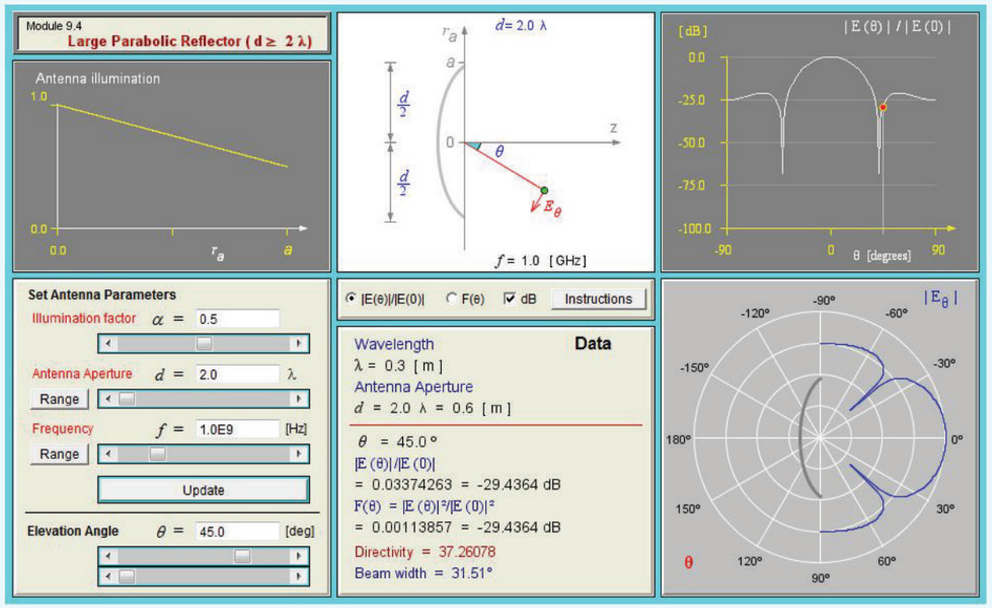
练习9-13：为了使用标量衍射来计算孔径天线的辐射场，必须满足什么条件？我们可以用它来计算光谱可见部分（$\lambda = 0.35$ 至 $0.7\ \mu$m）中眼睛瞳孔（$d \approx 0.2$ cm）的方向图吗？ $\lambda = 0.5\ \mu$m 处眼睛方向图的波束宽度是多少？

答案：$\beta \approx \lambda/d = 2.5 \times 10^{-4}$ rad $= 0.86'$ (角分，$60' = 1°$)。 （参见EM。）

9-9 天线阵列

AM 广播服务在 535 至 1605 kHz 频段运行。他们使用的天线是沿着高塔安装的垂直偶极子。天线的高度范围从 $\lambda/6$ 到 $5\lambda/8$，具体取决于所需的操作特性和其他考虑因素。它们的物理高度从 46 m（150 英尺）到 274 m（900 英尺）不等； 1 MHz 处的波长（大约位于 AM 波段的中间）为 300 m。因为单个偶极子辐射的场在水平面内是均匀的

Module 9.4 Large Parabolic Reflector For any specified reflector diameter d (where $d \geq 2\lambda$) and illumination taper factor α, this module displays the pattern of the radiated field and computes the associated beamwidth and directivity.



(as discussed in Sections 9-1 and 9-3), it is not possible to direct the horizontal pattern along specific directions of interest, unless two or more antenna towers are used simultaneously. Directions of interest may include cities serviced by the AM station, and directions to avoid may include areas serviced by another station operating at the same frequency (thereby avoiding interference effects). When two or more antennas are used together, the combination is called an ***antenna array***.

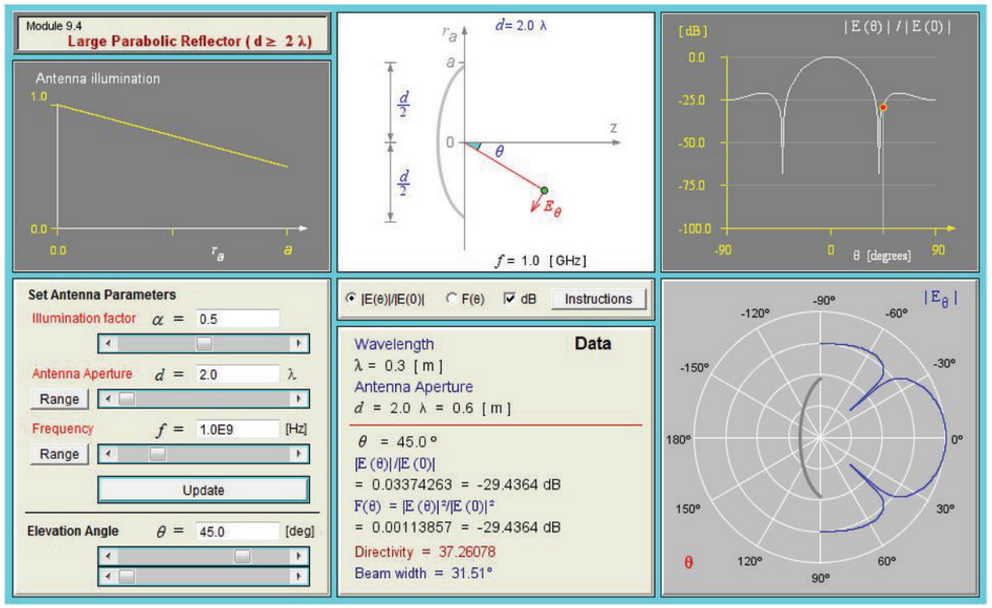
The AM broadcast antenna array is only one example of the many antenna arrays used in communication systems and radar applications. Antenna arrays provide the antenna designer the flexibility to obtain high directivity, narrow beams, low side lobes, steerable beams, and shaped antenna patterns starting from very simple antenna elements. **Figure 9-25** shows a very large radar system consisting of a transmitter array composed of 5,184 individual dipole antenna elements and a receiver array composed of 4,660 elements. The radar system, part of the Space Surveillance Network operated by the U.S. Air Force, operates at 442 MHz and transmits a combined peak power of 30 MW!

Although an array need not consist of similar radiating elements, most arrays actually use identical elements, such as dipoles, slots, horn antennas, or parabolic dishes. The antenna elements comprising an array may be arranged in various configurations, but the most common are the linear one-dimensional configuration—wherein the elements are arranged along a straight line—and the two-dimensional lattice configuration in which the elements sit on a planar grid. The desired shape of the far-field radiation pattern of the array can be synthesized by controlling the relative amplitudes of the array elements' excitations.

► Also, through the use of electronically controlled solid-state phase shifters, the beam direction of the antenna array can be steered electronically by controlling the relative phases of the array elements. ◄

This flexibility of the array antenna has led to numerous applications, including ***electronic steering*** and ***multiple-beam generation***.

模块 9.4 大型抛物面反射器 对于任何指定的反射器直径 d（其中 d ≥ 2λ）和照明锥度系数 α，该模块显示辐射场的图案并计算相关的波束宽度和方向性。



（如第 9-1 和 9-3 节中讨论的），除非同时使用两个或更多天线塔，否则不可能沿着特定的感兴趣方向引导水平方向图。感兴趣的方向可能包括 AM 电台服务的城市，而要避开的方向可能包括以相同频率运行的另一个电台服务的区域（从而避免干扰影响）。当两个或多个天线一起使用时，这种组合称为天线阵列。

AM 广播天线阵列只是通信系统和雷达应用中使用的众多天线阵列之一。天线阵列为天线设计者提供了从非常简单的天线元件开始获得高方向性、窄波束、低旁瓣、可操纵波束和成形天线方向图的灵活性。图9-25显示了一个非常大的雷达系统，其中包括由5,184个独立偶极天线元件组成的发射器阵列和由4,660个元件组成的接收器阵列。该雷达系统是美国空军运营的太空监视网络的一部分，工作频率为 442 MHz，发射的综合峰值功率为 30 MW！

尽管阵列不需要由类似的辐射元件组成，但大多数阵列实际上使用相同的元件，例如偶极子、缝隙、喇叭天线或抛物面天线。组成阵列的天线元件可以以各种配置排列，但最常见的是线性一维配置（其中元件沿直线排列）和二维晶格配置（其中元件位于平面网格上）。可以通过控制阵列元件激励的相对幅度来合成阵列远场辐射图的所需形状。

► 此外，通过使用电子控制的固态移相器，可以通过控制阵列元件的相对相位来电子地控制天线阵列的波束方向。 ◄

阵列天线的这种灵活性带来了众多应用，包括电子转向和多波束生成。



Figure 9-25 The AN/FPS-85 Phased Array Radar Facility in the Florida panhandle is near the city of Freeport. A several-mile no-fly zone surrounds the radar installation as a safety concern for electroexplosive devices, such as ejection seats and munitions that are carried on military aircraft.

The purpose of this and the next two sections is to introduce the reader to the basic principles of array theory and design techniques used in shaping the antenna pattern and steering the main lobe. This presentation is confined to the one-dimensional linear array with equal spacing between adjacent elements.

A linear array of N identical radiators is arranged along the z axis, as shown in **Fig. 9-26**. The radiators are fed by a common oscillator through a branching network. In each branch, an attenuator (or amplifier) and phase shifter are inserted in series to control the amplitude and phase of the signal feeding the antenna element in that branch.

In the far-field region of any radiating element, the ***element electric-field intensity*** $\widetilde{E}_e(R,\theta,\phi)$ may be expressed as a product of two functions: the spherical propagation factor e^{-jkR}/R, which accounts for the dependence on the range R, and $\widetilde{f}_e(\theta,\phi)$, which accounts for the directional dependence of the element's electric field. Thus, for an isolated element, the radiated field is

$$\widetilde{E}_e(R,\theta,\phi) = \frac{e^{-jkR}}{R}\,\widetilde{f}_e(\theta,\phi), \tag{9.93}$$

and the corresponding power density S_e is

$$S_e(R,\theta,\phi) = \frac{1}{2\eta_0}|\widetilde{E}_e(R,\theta,\phi)|^2 = \frac{1}{2\eta_0 R^2}|\widetilde{f}_e(\theta,\phi)|^2. \tag{9.94}$$

Hence, for the array shown in **Fig. 9-26(b)**, the far-zone field due to element i at range R_i from observation point Q is

$$\widetilde{E}_i(R_i,\theta,\phi) = A_i\,\frac{e^{-jkR_i}}{R_i}\,\widetilde{f}_e(\theta,\phi), \tag{9.95}$$

where $A_i = a_i e^{j\psi_i}$ is a ***complex feeding coefficient*** representing the amplitude a_i and phase ψ_i of the excitation giving rise to $\widetilde{E}_i$, which is relative to a reference excitation. In practice, the excitation of one of the elements is used as reference. Note that R_i and A_i may be different for different elements in the array, but $\widetilde{f}_e(\theta,\phi)$ is the same for all of them because they are all identical; hence, they exhibit identical directional patterns.

The total field at the observation point $Q(R_0,\theta,\phi)$ is the sum of the fields due to the N elements:

$$\widetilde{E}(R_0,\theta,\phi) = \sum_{i=0}^{N-1}\widetilde{E}_i(R_i,\theta,\phi) = \left[\sum_{i=0}^{N-1}A_i\,\frac{e^{-jkR_i}}{R_i}\right]\widetilde{f}_e(\theta,\phi), \tag{9.96}$$

where R_0 denotes the range of Q from the center of the coordinate system, which is chosen to be at the location of the zeroth element. To satisfy the far-field condition given by Eq. (9.73) for an array of length $l = (N-1)d$, where d is the interelement spacing, the range R_0 should be sufficiently large to satisfy

$$R_0 \geq \frac{2l^2}{\lambda} = \frac{2(N-1)^2d^2}{\lambda}. \tag{9.97}$$

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图 9-25 佛罗里达州狭长地带的 AN/FPS-85 相控阵雷达设施靠近自由港市。雷达装置周围有一个数英里的禁飞区，这是军用飞机上携带的弹射座椅和弹药等电爆装置的安全问题。

本节和接下来两节的目的是向读者介绍阵列理论的基本原理以及用于塑造天线方向图和控制主瓣的设计技术。本演示仅限于相邻元素之间间距相等的一维线性阵列。

N个相同的辐射器沿z轴排列成线性阵列，如图9-26所示。辐射器由公共振荡器通过分支网络供电。在每个分支中，串联插入衰减器（或放大器）和移相器，以控制馈送到该分支中的天线元件的信号的幅度和相位。

在任何辐射元件的远场区域，元件电场强度 $\widetilde{E}_e(R、\theta、\phi)$ 可以表示为两个函数的乘积：球面传播因子 e^{-jkR}/R（解释对范围 R 的依赖性）和 $\widetilde{f}_e(\theta、\phi)$（解释元件电场的方向依赖性）。因此，对于孤立元件，辐射场为

$$\widetilde{E}_e(R,\theta,\phi) = \frac{e^{-jkR}}{R}\,\widetilde{f}_e(\theta,\phi), \tag{9.93}$$

相应的功率密度S_e为

$$S_e(R,\theta,\phi) = \frac{1}{2\eta_0}|\widetilde{E}_e(R,\theta,\phi)|^2 = \frac{1}{2\eta_0 R^2}|\widetilde{f}_e(\theta,\phi)|^2. \tag{9.94}$$

因此，对于图9-26(b)所示的阵列，远区场由于距观察点 Q 的范围 R_i 内的元素 i 为

$$\widetilde{E}_i(R_i,\theta,\phi) = A_i\,\frac{e^{-jkR_i}}{R_i}\,\widetilde{f}_e(\theta,\phi), \tag{9.95}$$

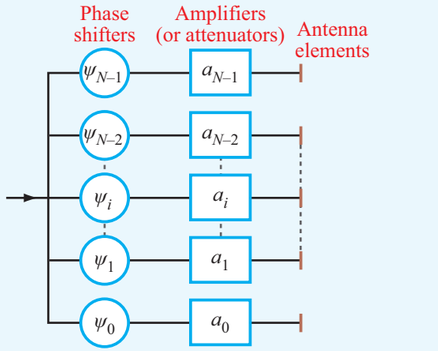
其中 $A_i = a_i e^{j\psi_i}$ 是一个复数馈送系数，表示产生 $\widetilde{E}_i$ 的激励的幅度 a_i 和相位 ψ_i，相对于参考激励。在实践中，以其中一种元素的激励作为参考。请注意，对于数组中的不同元素，R_i 和 A_i 可能不同，但 $\widetilde{f}_e(\theta、\phi)$ 对于所有元素来说都是相同的，因为它们都是相同的；因此，它们表现出相同的方向模式。

观察点 $Q(R_0,\theta,\phi)$ 处的总场是 N 个元素产生的场之和：

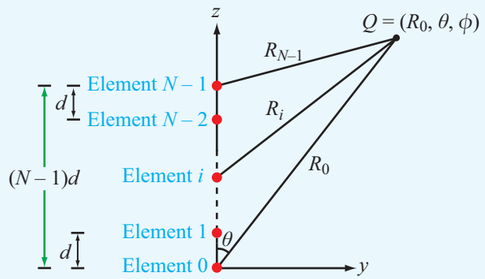
$$\widetilde{E}(R_0,\theta,\phi) = \sum_{i=0}^{N-1}\widetilde{E}_i(R_i,\theta,\phi) = \left[\sum_{i=0}^{N-1}A_i\,\frac{e^{-jkR_i}}{R_i}\right]\widetilde{f}_e(\theta,\phi), \tag{9.96}$$

其中R_0表示Q距坐标系中心的范围，该坐标系选择在第零个元素的位置。满足方程给出的远场条件。 (9.73) 对于长度为 $l = (N-1)d$ 的数组，其中 d 是元素间间距，范围 R_0 应足够大以满足

$$R_0 \geq \frac{2l^2}{\lambda} = \frac{2(N-1)^2d^2}{\lambda}. \tag{9.97}$$



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

Figure 9-26 Linear-array configuration and geometry.

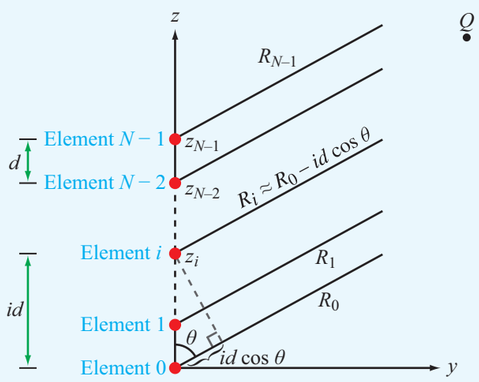


Figure 9-27 The rays between the elements and a faraway observation point are approximately parallel lines. Hence, the distance $R_i \approx R_0 - id\cos\theta$.

This condition allows us to ignore differences in the distances from Q to the individual elements as far as the magnitudes of the radiated fields are concerned. Thus, we can set $R_i = R_0$ in the denominator in Eq. (9.96) for all i. With regard to the phase part of the propagation factor, we can use the parallel-ray approximation given by

$$R_i \approx R_0 - z_i\cos\theta = R_0 - id\cos\theta, \qquad (9.98)$$

where $z_i = id$ is the distance between the ith element and the zeroth element (**Fig. 9-27**). Employing these two approximations in Eq. (9.96) leads to

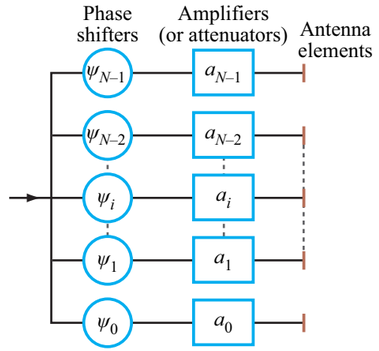
$$\widetilde{E}(R_0,\theta,\phi) = \widetilde{f}_e(\theta,\phi)\left(\frac{e^{-jkR_0}}{R_0}\right)\left[\sum_{i=0}^{N-1} A_i e^{jikd\cos\theta}\right], \qquad (9.99)$$

and the corresponding array-antenna power density is given by

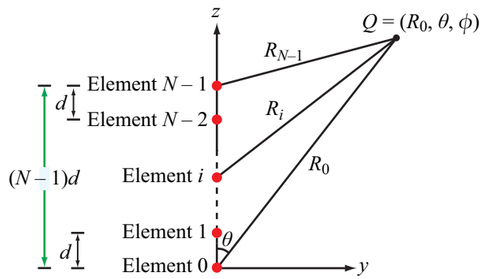
$$\begin{aligned} S(R_0,\theta,\phi) &= \frac{1}{2\eta_0}|\widetilde{E}(R_0,\theta,\phi)|^2 \\ &= \frac{1}{2\eta_0 R_0^2}|\widetilde{f}_e(\theta,\phi)|^2\left|\sum_{i=0}^{N-1} A_i e^{jikd\cos\theta}\right|^2 \\ &= S_e(R_0,\theta,\phi)\left|\sum_{i=0}^{N-1} A_i e^{jikd\cos\theta}\right|^2, \end{aligned} \qquad (9.100)$$

where use was made of Eq. (9.94). This expression is a product of two factors. The first factor, $S_e(R_0,\theta,\phi)$, is the power density of the energy radiated by an individual element, and the second, called the ***array factor***, is a function of the positions of the individual elements and their feeding coefficients but not a function of the specific type of radiators used.

► The array factor represents the far-field radiation intensity of the N elements—had the elements been isotropic radiators. ◄



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

图 9-26 线性阵列配置和几何结构。

就辐射场的大小而言，这种条件允许我们忽略 Q 到各个元件的距离差异。因此，我们可以将 $R_i = R_0$ 设置为式（1）中的分母。(9.96) 对于所有 i。对于传播因子的相位部分，我们可以使用下式给出的平行射线近似：

$$R_i \approx R_0 - z_i\cos\theta = R_0 - id\cos\theta, \qquad (9.98)$$

其中 $z_i = id$ 是第 i 个元素和第 0 个元素之间的距离（图 9-27）。采用这两个近似

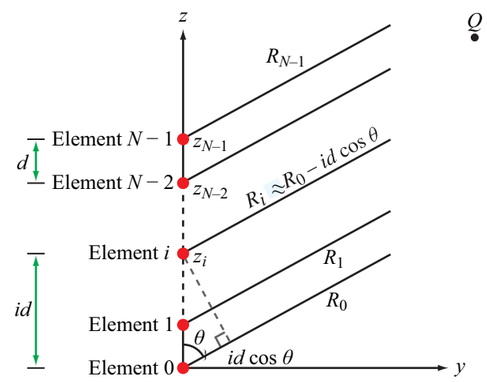


图9-27 单元与远处观察点之间的光线近似为平行线。因此，距离 $R_i \approx R_0 - id\cos\theta$。

方程式中的蒸发散(9.96) 导致

$$\widetilde{E}(R_0,\theta,\phi) = \widetilde{f}_e(\theta,\phi)\left(\frac{e^{-jkR_0}}{R_0}\right)\left[\sum_{i=0}^{N-1} A_i e^{jikd\cos\theta}\right], \qquad (9.99)$$

相应的阵列天线功率密度由下式给出

$$\begin{aligned} S(R_0,\theta,\phi) &= \frac{1}{2\eta_0}|\widetilde{E}(R_0,\theta,\phi)|^2 \\ &= \frac{1}{2\eta_0 R_0^2}|\widetilde{f}_e(\theta,\phi)|^2\left|\sum_{i=0}^{N-1} A_i e^{jikd\cos\theta}\right|^2 \\ &= S_e(R_0,\theta,\phi)\left|\sum_{i=0}^{N-1} A_i e^{jikd\cos\theta}\right|^2, \end{aligned} \qquad (9.100)$$

其中使用了方程式。（9.94）。该表达式是两个因素的产物。第一个因子 $S_e(R_0、\theta、\phi)$ 是单个元件辐射的能量的功率密度，第二个因子称为阵列因子，是单个元件的位置及其馈电系数的函数，但不是所使用的特定类型辐射器的函数。

► 阵列因子代表 N 个元件的远场辐射强度（假设这些元件是各向同性辐射器）。 ◄